

Executable Jars

Executable Jars with Eclipse

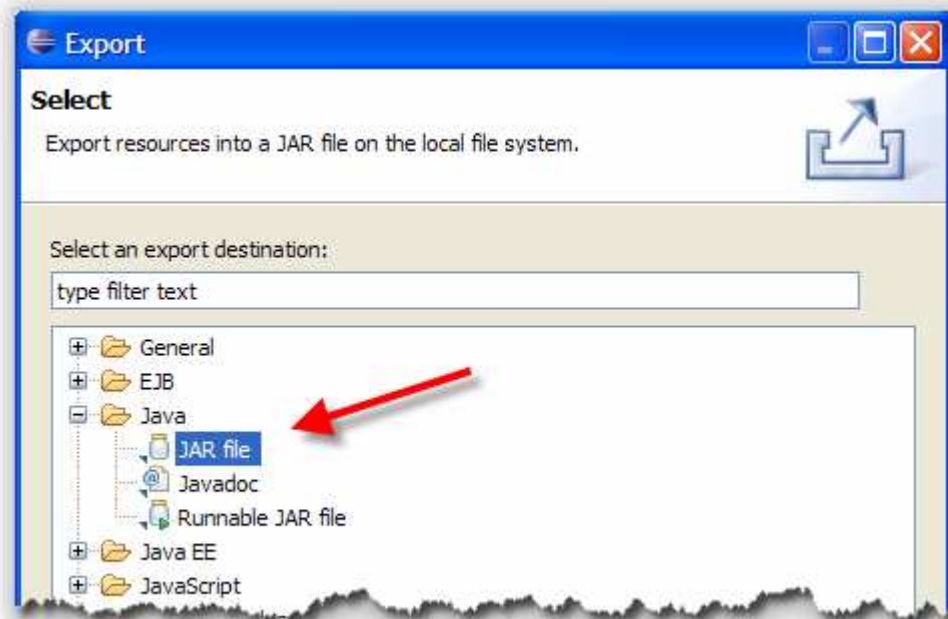
An executable **jar** (Java Archive) is simply a zip file that contains all of the classes and resources for an application, along with a **manifest** file that tells the Java interpreter where to find the **main()** method to start the program running. An executable **jar** allows you to package all of the various pieces of your program into an easy to distribute format.

If you have a GUI program packaged as an executable **jar**, you can usually just double-click it, like you would any executable, to start it running. For console-mode programs, like **Nim**, you'll run it by using the Java interpreter from the command line, along with the **-jar** switch, like this:

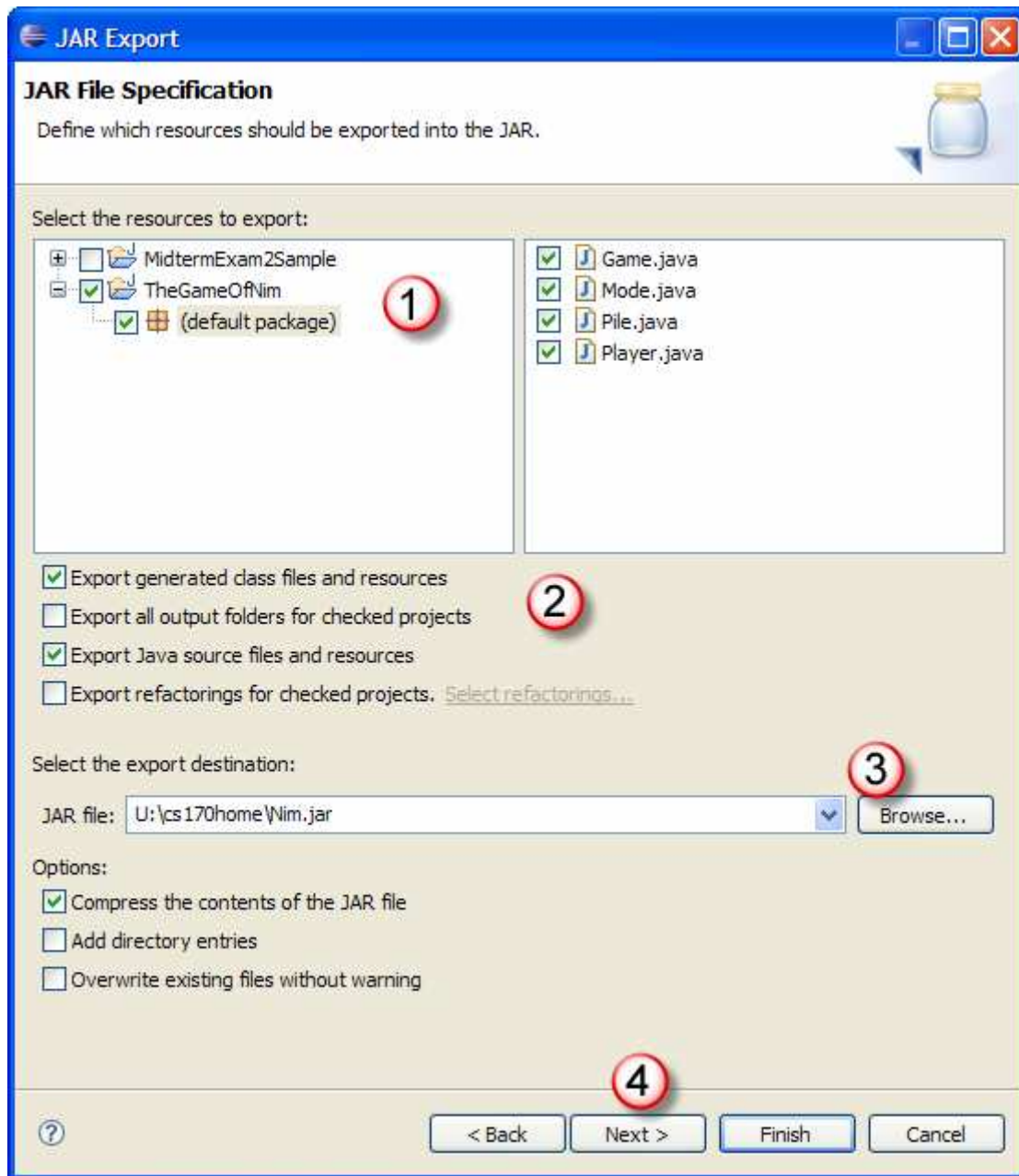
```
java -jar Nim.jar
```

Creating the jar with Eclipse

To create a **jar** file, you first choose **Export** from the **File** menu, after you've completed and debugged your application. On the **Select** dialog, expand the Java folder and select **JAR File** as shown here (*not* the Runnable JAR File):



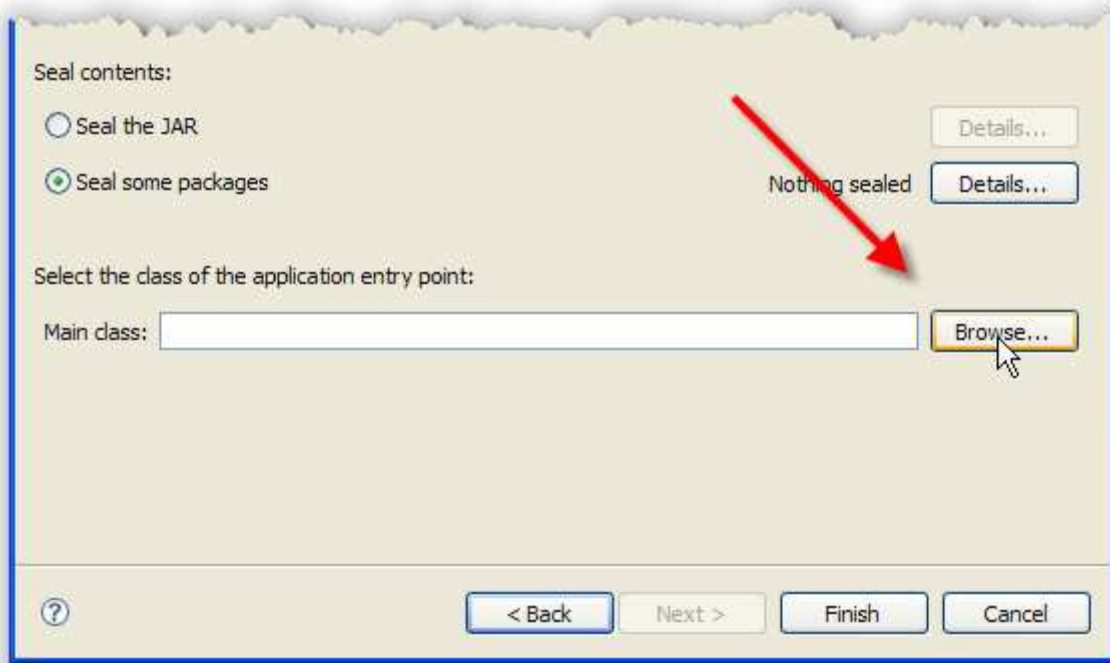
Click the **Next** button at the bottom of the dialog and you'll be taken to the **JAR Export** screen.



On the **JAR Export** screen you need to first select the project that you want to export. Then, make sure that the **Export generated class files and resources** checkbox is selected (2). To include the Java source code as well, select **Export Java source files and resources** also.

To select the export destination, either type a name into the **JAR file:** text box or click the **Browse** button appearing to its right and navigate to the directory where you want to store your file. Finally, click the **Next** button to go on to the next screen: **JAR Packaging Options**. There is nothing you need to change on this screen, so just click **Next** once again to go to the **JAR Manifest Generation** screen which I've reproduced on the next page.

When the **JAR Manifest Generation** screen appears, locate the **Main class** text box and click the **Browse** button that appears at its right as shown here:



When the **Select Main Class** dialog appears (on the right), select the class containing your **main()** method and click the **OK** button.

Click the **Finish** button on the **JAR Manifest Generation** screen to finish the process.

